

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Phase 1, Multicenter, Single-arm, Dose-escalation Study of CC-97540 (BMS-986353), CD19-Targeted NEX-T Chimeric Antigen Receptor (CAR) T Cells, Evaluating Safety and Tolerability in Participants With Autoimmune Neurological Diseases: Relapsing Forms of Multiple Sclerosis (RMS), Progressive Forms of Multiple Sclerosis (PMS), or Refractory Myasthenia Gravis (MG).

Brief Study Title: A Study to Evaluate the Safety, Tolerability, Efficacy, and Drug Levels of CC-97540 in Participants With Relapsing Forms of Multiple Sclerosis, Progressive Forms of Multiple Sclerosis or Refractory Myasthenia Gravis (MG) (Breakfree-2). **Short Title/Acronym:** Breakfree-2.

Protocol Number: CA061-1006.

NCT Number: NCT06220201. **Posting ID:** 343190604473177577. **Trial Status:** RECRUITING (Currently recruiting participants). **Sponsor/Company Name:** Bristol Myers Squibb (Juno Therapeutics, Inc.).

Investigational Product: CC-97540 (BMS-986353), an investigational CD19-directed CAR T cell therapy.

Phase of Development: Phase 1.

Medical Monitor: Bristol Myers Squibb Clinical Operations Lead.

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and dose-limiting toxicities of a single infusion of CC-97540 (BMS-986353) in patients with RMS, PMS, or MG, and to determine the recommended phase 2 dose (RP2D). **Secondary Objectives:** To evaluate the efficacy, pharmacokinetics, and drug levels of CC-97540.

2.2 Background & Rationale Approved therapies for relapsing and progressive multiple sclerosis have a limited effect on disease progression and possess concerning long-term toxicities. Novel chimeric antigen receptor (CAR) T cell therapies, such as CC-97540 (BMS-986353), target CD19 to deplete B cells deeply in the central nervous system and periphery. This mechanism aims to exert anti-inflammatory and immunomodulatory effects, inducing an immune reset that could possibly lead to durable, treatment-free disease control in severe autoimmune neurological diseases.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional. **Control Type:** Single-arm (Dose-escalation). **Blinding:** Open-label. **Randomization:** N/A.

3.2 Planned Enrollment & Duration Target **N**: 120 participants. **Number of Sites:** Multicenter (United States and 5 other countries). **Estimated Study Start:** 27-Mar-2024. **Estimated Primary Completion:** 14-Jul-2027.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 60 years. 2. **RMS (Cohort 1):** Expanded Disability Status Scale (EDSS) of ≥ 3.0 and ≤ 5.5 ; diagnosis of MS with relapsed/refractory MS or conversion to active secondary progressive MS (aSPMS), with disease worsening within 12 months prior to screening despite being on treatment with a high-efficacy DMT for at least 6 months. 3. **PMS (Cohort 2):** EDSS ≥ 3.0 and ≤ 6.0 ; diagnosis of primary progressive MS (PPMS) that is treatment-resistant or inactive secondary progressive MS (iSPMS). 4. **Refractory MG (Cohort 3):** - MGFA classification of II-IV at screening. - Documentation of autoantibodies against AChR or MuSK (historical or at Screening). - Refractory disease defined as disease activity on at least 2 immunosuppressants (including steroids, NSIs, or biologics). - Has had prior thymectomy only if indicated according to current guidelines.

4.2 Exclusion Criteria 1. Cohorts 1 and 2: Participants that cannot complete the 9-Hole Peg Test (9-HPT) in at least 1 hand in < 240 seconds unless extenuating medical conditions unrelated to MS prohibit this. 2. Participants that cannot perform a Timed 25-Foot Walk Test (T25FWT) in < 150 seconds. 3. Presence of other confounding peripheral nervous system disorders or conditions that may impact muscle strength or cause weakness (e.g., myositis, stroke, chronic inflammatory demyelinating polyradiculoneuropathy, Lambert-Eaton myasthenic syndrome).

5. Intervention & Treatment Plan

5.1 Investigational Regimen & Pre-conditioning **Pre-conditioning Therapy:** Cyclophosphamide (ATC Code: L01AA01; Trade names: Cytoxan, Endoxana). **Dosage/Frequency:** Single infusion of CC-97540 (BMS-986353) CAR T cells in a dose-escalation cohort approach following pre-conditioning. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Single administration followed by long-term post-infusion safety follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for monitoring and managing treatment-emergent

adverse events. **Prohibited:** Concomitant use of disease-modifying or immunosuppressive therapies that conflict with CAR T cell survival, unless specifically treating protocol-defined toxicity (e.g., Cytokine Release Syndrome).

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Days -28 to 0	Informed Consent, Eligibility Criteria Check, Baseline EDSS/9-HPT/T25FWT.
Baseline	Day 0	Cyclophosphamide Pre-conditioning, CC-97540 Infusion.
Interim	Post-Infusion	Dose-limiting toxicity (DLT) evaluation, CRS grading, pharmacokinetic assays (Cmax, AUC).
End of Study	Up to 2027	Long-term disease progression checks, final clinical monitoring, resolution of AESIs.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Evaluating safety by tracking:
1. Number of participants with adverse events (AEs). 2. Number of participants with serious adverse events (SAEs). 3. Number of participants with adverse events of special interest (AESIs, such as CRS). 4. Dose-limiting toxicities (DLTs) and determination of Recommended Phase 2 Dose (RP2D). **Measurement Method:** Clinical assessment utilizing the ASTCT Consensus Grading System for specific toxicities.

7.2 Secondary & Exploratory Endpoints - Number of participants meeting no evidence of disease activity (NEDA) criteria. - Number of participants with confirmed disability progression (CDP) and confirmed disability improvement (CDI) per Expanded Disability Status Scale (EDSS). - Annualized relapse rate. - Pharmacokinetics including maximum observed blood concentration (Cmax), area under the blood concentration-time curve (AUC), and time to last measurable CAR concentration (Tlast).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring including specialized grading of treatment-related AEs (TRAEs) like CRS and neurologic toxicities. **Serious Adverse Events (SAEs):** Reported continuously. Focus during the acute post-infusion window centers on identifying DLTs. **Data Monitoring Committee (DMC):** Review of safety and tolerability by standard phase 1 dose-escalation committees to transition smoothly between dose levels.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for AE/SAE/DLT evaluations; ITT/Enrolled populations for longitudinal clinical efficacy metrics like EDSS changes and annualized relapse rates. **Sample Size**

Justification: $N=120$ participants target optimally powers the dose-escalation phases to confidently evaluate safety and recommend an RP2D. **Handling of Missing Data:** Standard handling aligned with phase 1 cellular therapy guidelines, generally utilizing actual observed values during DLT windows.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Constant safety, toxicity, and dose-escalation clinical checks managed by the sponsor across testing sites.

Audit Trail: Stringent QA/QC measures regarding the NEX-T CAR-T cell manufacturing process, chain of identity, and clinical data reporting.

11. Study Identifiers & Governance

Primary ID: CA061-1006 **Registry Number:** NCT06220201 **Posting ID:** 343190604473177577 **Trial Status:** RECRUITING **Sponsor/Lead**

Organization: Bristol Myers Squibb / Juno Therapeutics, Inc.

Study Title: Breakfree-2 **Official Regulatory Title:** A Phase 1, Multicenter, Single-arm, Dose-escalation Study of CC-97540 (BMS-986353), CD19-Targeted NEX-T Chimeric Antigen Receptor (CAR) T Cells, Evaluating Safety and Tolerability in Participants With Autoimmune Neurological Diseases: Relapsing Forms of Multiple Sclerosis (RMS), Progressive Forms of Multiple Sclerosis (PMS), or Refractory Myasthenia Gravis (MG)

12. Clinical Framework (Autoimmune Neurological Specific)

Primary Conditions: Multiple Sclerosis (RMS, PMS) & Refractory Myasthenia Gravis (MG) **Preferred UMLS Name:** Autoimmune Neurological Disease **Semantic Type:** Disease or Syndrome **Diagnosis Threshold:** EDSS ≥ 3.0 and ≤ 6.0 ; or treatment-resistant activity on ≥ 2 immunosuppressants **Medical Methodology:** CD19-Targeted NEX-T Chimeric Antigen Receptor (CAR) T Cell Therapy **Optimization Target:** Deep B-cell depletion in the central nervous system and periphery to achieve an immune reset

13. Study Procedures & Methodology

Management Pathway: CAR-T Pre-conditioning (Cyclophosphamide) and Infusion Clinical Pathway **Education Protocol (HCP):** Dose-limiting toxicity (DLT) identification, cytokine release syndrome (CRS) grading, and neurotoxicity management **Education Protocol (Patient):** Acute post-infusion safety procedures, prolonged toxicity risks, and continuous trial-specific monitoring protocols **Quality Audit Mechanism:** NEX-T manufacturing QC / safety incidence logging

14. Observation & Follow-Up Schedule

Total Duration: Until primary trial completion (projected 2027) spanning immediate DLT windows to long-term follow-up **Onsite Visit Cadence:** Close continuous monitoring initially post-infusion to observe for acute toxicities, stepping down to interval long-term efficacy monitoring **Remote Monitoring Frequency:** Routinely throughout the safety tracking period **Checklist Review Interval:** Regular tracking points for standardized motor and disability metrics (EDSS, 9-HPT, T25FWT)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				